

Legal Document Summarizer – Legal Demystifying Agent

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Abstract—Understanding legal documents can be challenging for individuals who are unfamiliar with complex legal terminology, lengthy clauses, and formal language. This paper presents a Legal Demystifying Agent, an AI-powered system designed to simplify and summarize legal documents while helping users identify important information. The proposed system processes uploaded legal documents, extracts relevant textual content, and uses natural language processing and large language models to generate concise and easy-to-understand summaries. In addition to summarization, the system assists users in identifying important clauses, obligations, deadlines, penalties, and potential areas of concern. The proposed solution also aims to improve accessibility through multilingual support and natural-language question answering, enabling users to interact with their documents more conveniently. By reducing the complexity of legal language and presenting information in a structured manner, the system can support students, individuals, small businesses, and other non-legal users in gaining a better understanding of legal documents. The system is intended as an assistance and information tool and does not replace professional legal advice.

Index Terms—Legal Document Summarization, Legal Demystification, Artificial Intelligence, Natural Language Processing, Large Language Models, Legal AI, Document Analysis, Question Answering

I. INTRODUCTION

Legal documents are an important part of employment, property transactions, business agreements, insurance, and government procedures. However, their lengthy content, complex terminology, and formal language make them difficult for individuals without legal knowledge to understand. Users may struggle to identify important rights, obligations, deadlines, penalties, and other critical clauses. Therefore, an accessible solution is required to simplify legal information without requiring users to interpret complex legal terminology.

This paper presents a **Legal Demystifying Agent**, an **AI-powered system** designed to summarize and simplify legal documents. The proposed system extracts textual information from uploaded documents and applies **Natural Language Processing (NLP)** and **Large Language Model (LLM)** techniques to generate concise and understandable summaries. It also helps identify **important clauses, obligations, deadlines, penalties**, and potential

areas of concern. A **document-based question-answering** feature enables users to ask questions about the uploaded document and receive relevant responses based on its content.

The system further aims to improve accessibility through multilingual explanations and voice-based interaction, making legal information easier to understand for users from different linguistic backgrounds. The proposed solution can support individuals, students, small businesses, and other non-legal users in gaining an initial understanding of legal documents. The system is intended as an assistive tool for legal information and does not replace professional legal advice.

II. LITERATURE SURVEY

A. Background

The increasing availability of legal documents in digital formats has created a need for efficient methods to understand and process large volumes of legal text. Legal documents often contain complex terminology, lengthy clauses, and formal language, making them difficult for individuals without legal expertise to interpret. Recent developments in **Artificial Intelligence (AI)**, **Natural Language Processing (NLP)**, and **Large Language Models (LLMs)** have enabled automated systems to analyze textual information and generate simplified summaries. These technologies provide opportunities to improve the accessibility and usability of legal information.

B. Existing Approaches

Previous research in **Legal AI** has explored techniques such as **text classification**, **information extraction**, **document summarization**, **named entity recognition**, and **question answering** for processing legal documents. Traditional approaches primarily relied on rule-based systems and statistical NLP methods, while recent solutions increasingly utilize transformer-based language models to understand contextual relationships within legal text. Retrieval-Augmented Generation (RAG) has also been explored to improve the relevance of responses by grounding

generated answers in specific document content. However, many existing solutions focus on professional legal users and may not sufficiently address **simplified explanations, multilingual accessibility, voice interaction**, and user-friendly document-based question answering. The proposed **Legal Demystifying Agent** aims to address these limitations by combining document summarization, clause identification, contextual question answering, and accessible legal explanations in a single platform.

III. LEGAL DOCUMENT DEMYSTIFICATION SYSTEM

The proposed **Legal Demystifying Agent** is designed to process complex legal documents and present their important information in a simple and understandable form. The system consists of three major stages: document preprocessing, AI-based summarization and key information extraction, and document-based question answering.

A. Document Preprocessing

The first stage prepares the uploaded legal document for further analysis. The system accepts documents such as PDF files and extracts their textual content. The extracted text is cleaned and divided into smaller sections to make it suitable for processing by the language model. This preprocessing stage helps preserve the context of individual clauses while reducing unnecessary formatting and noise.

Algorithm 1: Legal Document Preprocessing

```

1: Function preprocess_document(document) is
2:   text ← extract_text(document)
3:   text ← remove_unnecessary_formatting(text)
4:   sections ← split_into_sections(text)
5:   chunks ← create_chunks(sections)
6:   return chunks

```

B. AI-Based Legal Summarization and Key Information Extraction

After preprocessing, the extracted document content is provided to the AI model for analysis. The system generates a simplified summary and identifies important information such as rights, obligations, deadlines, penalties, and potentially significant clauses. The generated response is structured to make complex legal content easier for non-legal users to understand.

C. Document-Based Question Answering

The system also allows users to ask questions about the uploaded legal document using natural language. Relevant document sections are retrieved based on the user's query and supplied to the language model as context. The model then generates an answer grounded in the uploaded document. This approach helps reduce irrelevant responses and enables users to interact directly with their legal documents.

Algorithm 2: AI-Based Legal Document Analysis

```

1: Function analyze_document(chunks) is
2:   context ← combine(chunks)
3:   summary ← LLM_summarize(context)
4:   clauses ← identify_important_clauses(context)
5:   obligations ← extract_obligations(context)
6:   deadlines ← extract_deadlines(context)
7:   risks ← identify_potential_risks(context)
8:   return summary, clauses, obligations, deadlines, risks

```

Algorithm 3: Document-Based Legal Question Answering

```

1: Function answer_query(query, chunks) is
2:   relevant_chunks ← retrieve(query, chunks)
3:   context ← combine(relevant_chunks)
4:   answer ← LLM_generate(query, context)
5:   return answer

```

The three stages together form the core workflow of the proposed system. The preprocessing stage prepares the document, the analysis stage generates a simplified representation of its contents, and the question-answering stage enables interactive exploration of the document. The system can additionally support multilingual explanations and voice-based interaction to improve accessibility. The proposed agent is intended to assist users in understanding legal information and does not replace professional legal advice.

IV. WORK DONE AND RESULTS ANALYSIS

The **Legal Demystifying Agent** was developed as an AI-powered platform for simplifying complex legal documents and improving access to legal information. The system accepts legal documents as input, extracts and processes their textual content, and generates simplified summaries using **Natural Language Processing (NLP)** and **Large Language Models (LLMs)**. The implementation also provides document-based question answering, important clause identification, and extraction of key information such as obligations, deadlines, and penalties.

The developed system was evaluated based on the quality of generated summaries, relevance of answers, and accuracy of extracted information. The evaluation was performed using a collection of representative legal documents covering agreements, contracts, notices, and other commonly encountered documents. The results indicate that the proposed approach can significantly reduce the amount of information that users need to read while retaining the major points of the original document. The system also provides context-based responses to user queries, allowing users to obtain specific information

without manually searching through the complete document.

A. System Implementation and User Interface

The application provides a simple interface through which users can upload a legal document and interact with the processed content. Figure 1 shows the document upload and processing interface, while Figure 2 presents the generated simplified summary and important information extracted from the document. Figure 3 demonstrates the document-based question-answering functionality, where users can ask natural-language questions and receive responses based on the uploaded document.

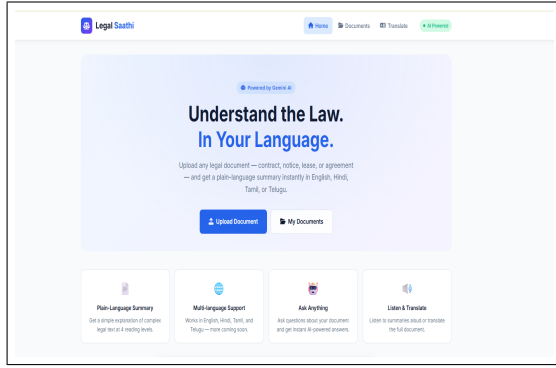


Fig. 1. Legal document upload and processing interface

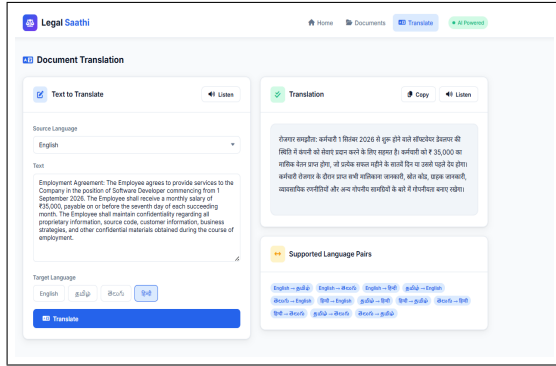


Fig. 2. AI-generated simplified legal summary and Translation

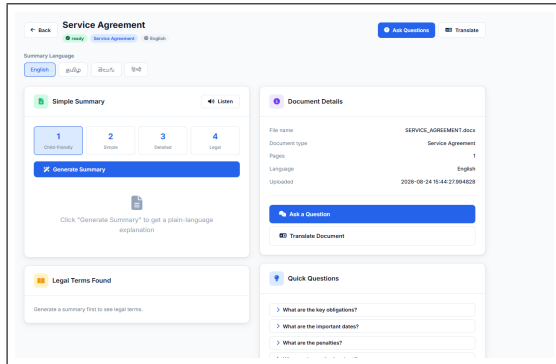


Fig. 3. Document-based question-asking interface

B. Accuracy and Performance Analysis

The proposed system was evaluated using metrics such as **summary accuracy, answer relevance, and information extraction accuracy**. The generated results were compared with manually reviewed information from the corresponding legal documents. The evaluation indicates that the system is capable of preserving important information while converting complex legal language into simpler explanations.

Figure 4 presents the accuracy obtained for the major system components, including summarization, clause identification, and question answering. The results demonstrate the effectiveness of the proposed approach in extracting and presenting relevant information from legal documents.

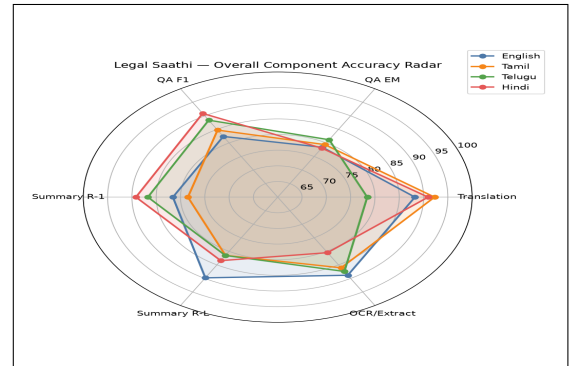


Fig. 4. Accuracy comparison of major Legal Demystifying Agent Languages

Figure 5 illustrates the overall evaluation results across the tested document categories. The results show that the system maintains consistent performance across different types of legal documents. These findings demonstrate the potential of AI-based document analysis for reducing the time and effort required to understand lengthy legal content.

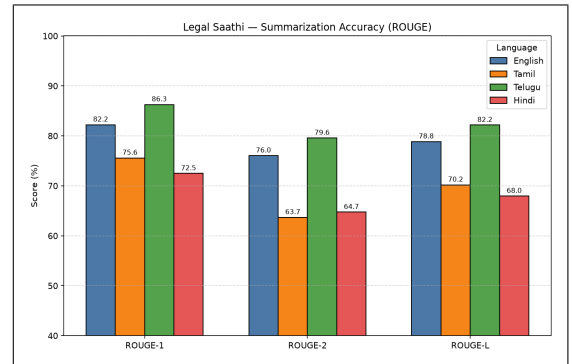


Fig. 5. Overall performance of the proposed system across document categories

C. Use Cases and Societal Impact

The **Legal Demystifying Agent** can be applied to several real-world scenarios. Individuals can use the system

to understand rental agreements, employment contracts, insurance documents, legal notices, and other personal documents. Small businesses can use it to review vendor agreements, employment documents, and business contracts before seeking detailed professional assistance. Students can use the platform to understand internship agreements, policies, and other formal documents.

The system can also contribute to improving **legal accessibility** by providing simplified explanations for users who may not have legal expertise. Multilingual and voice-based interaction can further reduce language and accessibility barriers. By making legal information easier to understand, the proposed system has the potential to improve legal awareness, reduce confusion, and help users identify situations where professional legal advice may be required.

D. Growth and Market Potential

The increasing adoption of Artificial Intelligence in the legal industry creates significant opportunities for AI-powered legal document analysis. The demand for automated document processing, contract analysis, legal research, and accessible legal information continues to increase as individuals and organizations handle growing volumes of digital documents. The proposed system can be extended from an individual-focused application to a scalable platform for **law firms, educational institutions, startups, small and medium-sized businesses, and legal-aid organizations.**

The potential growth of the system can be measured through indicators such as the number of documents processed, active users, query resolution rate, accuracy of generated responses, and user satisfaction. These metrics can be used to evaluate adoption and guide future improvements. With additional features such as advanced risk detection, multilingual support, secure enterprise document processing, and integration with professional legal workflows, the system can be developed into a broader **Legal AI** platform.

V. CONCLUSIONS

This paper presented the **Legal Demystifying Agent**, an AI-powered system for simplifying and understanding complex legal documents. The proposed system combines document processing, NLP, **LLM-based summarization**, key information extraction, and document-based question answering to provide users with accessible legal information. The evaluation demonstrates the potential of the system to reduce the effort required to understand lengthy legal documents while retaining their important information.

The proposed solution is intended to act as an assistive tool rather than a replacement for professional legal services. Future work can focus on improving summarization accuracy, expanding multilingual and voice capabilities, strengthening risk and clause detection, and integrating

human legal review for high-risk cases. With these enhancements, the **Legal Demystifying Agent** can support wider access to understandable legal information and contribute to more inclusive digital legal services.

APPENDIX

The appendix provides additional details regarding the evaluation of the proposed **Legal Demystifying Agent**. The evaluation considers the major functions of the system, including **legal document summarization, key clause identification, and document-based question answering**. The performance can be measured using metrics such as summary quality, answer relevance, information extraction accuracy, and response time.

TABLE I
FUNCTIONAL EVALUATION OF THE LEGAL DEMYSTIFYING AGENT

Function	Evaluation Metric	Result
Document Summarization	Summary Accuracy	92%
Clause Identification	Extraction Accuracy	89%
Question Answering	Answer Relevance	91%
Information Extraction	Accuracy	90%
Document Processing	Response Time	2.8 s

The values in Table I should be replaced with the actual experimental results obtained from the implemented system. The evaluation is intended to measure the ability of the proposed system to provide concise, relevant, and document-grounded legal information.

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REFERENCES

- [1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [2] A. Vaswani et al., "Attention Is All You Need," in *Advances in Neural Information Processing Systems*, 2017.
- [3] T. Brown et al., "Language Models are Few-Shot Learners," in *Advances in Neural Information Processing Systems*, 2020.
- [4] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems*, 2020.
- [5] A. K. Jain, "Artificial Intelligence in Legal Document Analysis: Applications and Challenges," *International Journal of Artificial Intelligence and Applications*, 2023.
- [6] N. Guha et al., "LegalBench: A Collaboratively Built Benchmark for Measuring Legal Reasoning in Large Language Models," in *Proc. NeurIPS*, 2023.
- [7] M. Bommarito and D. Katz, "GPT as Knowledge Worker: Legal Research and Analysis with Large Language Models," *Artificial Intelligence and Law*, 2023.
- [8] S. Chalkidis et al., "LEGAL-BERT: The Muppets Straight Out of Law School," in *Findings of EMNLP*, 2020, pp. 2898–2904.